PASSWORD CRACKING REPORT

John the Ripper & Networkwalks Password Cracking

W3-PM-FINAL | CYBERSECURITY | NETWORKWALKS

Pentester Name	Shan Rasool
Program / Batch	B083 – Networkwalks
Modules completed	W3-PM1 (Password Cracking with John the Ripper) W3-PM2 (Networkwalks Password Cracking Tools)
Client / Target	Protected PDF files supplied as part of the Networkwalks educational project
Permission secured?	Yes – all activities were carried out inside the assigned educational scope, using only the lab materials supplied for this exercise.
Phases covered	Phase 1: Hash Extraction & Preparation Phase 2: Password Recovery Phase 3: Credential Verification & Evidence Documentation

1. Liability Disclaimer

All of the activities described in this report were carried out strictly inside the assigned educational cybersecurity scope, and only against the protected PDF files that were provided or authorised for this training exercise. Nothing in this document was run against a system, account or file that falls outside that scope. All material in this report exists purely for the purposes of education and research, and none of it should be used to break any law.

The instructor, the authors and Networkwalks accept no responsibility for any misuse of the knowledge contained in this report. Responsibility for any action taken outside an authorised scope rests entirely with the person who performs it, and it is worth stressing that unauthorised access to a system or file can amount to a criminal offence even in cases where no damage results.

2. Introduction

This report documents the practical work I completed during Week 3 of the Cybersecurity & Ethical Hacking program at Networkwalks. The focus of the week was password recovery from password-protected PDF files, approached from two different angles: first with the well-known command-line tool John the Ripper, driven through its Johnny graphical interface, and second with the browser-based Networkwalks Hash Calculator and Password Cracker.

Both exercises were completed inside a controlled educational environment using PDF files supplied specifically for this lab. Every stage of the process, from extracting a crackable hash out of a protected PDF through to recovering its password and confirming that the file could be opened, is backed up by the evidence screenshots gathered in Section 6 of this report.

3. Tools Used

The table below lists each tool and file used while completing this week's practical exercises, along with a short note on what each one contributed to the overall workflow.

Tool / Material	Purpose
John the Ripper (JTR)	Command-line password-recovery tool used to perform the dictionary attack on the extracted PDF hash.
Johnny (GUI for JTR)	Graphical front end for John the Ripper used to load the hash file, start the attack and read back the recovered password.
Networkwalks Hash Calculator	Web-based utility used to convert a password-protected PDF into a crackable pdf2john / hashcat-compatible hash.
Networkwalks Password Cracker	Browser-based dictionary-attack tool used to test a wordlist against the extracted hash and recover the password.
Protected PDF challenge files	Two password-protected PDF files, PDF1 and PDF2, supplied for the lab and used as the target material for both exercises.

4. Activities Performed

4.1 W3-PM1 — Password Cracking with John the Ripper (JTR)

The first exercise walked through a complete JTR-based password-recovery workflow, built around six stages: confirming that the John the Ripper build was installed and working correctly, preparing the Johnny interface, extracting a crackable hash from the target file (PDF1), loading that hash into Johnny, running the attack to recover the password, and finally opening PDF1 to confirm that the recovered password was correct.

The process started by running john.exe --list=build-info from PowerShell, which printed the installed version (1.9.0-jumbo-1), the supported hash formats and the cryptographic libraries available to the tool. This step matters because it confirms the tool is ready to use before any time is spent on an actual attack.

With the build confirmed, My Locked PDF1.pdf was uploaded to the Networkwalks Hash Calculator, which extracted a pdf hash in the pdf2john / hashcat-compatible format. That hash was then imported into Johnny, where it appeared as a single password-protected PDF entry with no password recovered yet. Running the attack from inside Johnny returned the password good-luck next to the matching hash, confirming that the dictionary attack had succeeded.

As a final check, the recovered password was used to open PDF1 directly. The file unlocked successfully and displayed a confirmation screen containing the first captured flag, nw{cybersecurity_flag_captured_2608}, which closed out this exercise.

4.2 W3-PM2 — Password Cracking with Networkwalks Tools

The second exercise repeated the same overall idea — extract a hash, recover a password, verify the result — but used the Networkwalks web tools from start to finish instead of John the Ripper. The target this time was a second protected file, My Locked PDF2.pdf.

The file was uploaded to the Networkwalks Hash Calculator, which again produced a crackable pdf hash and additionally reported the PDF's revision, version and key length, giving a little more context on how the file had been encrypted. That hash was then pasted into the Networkwalks Password Cracker, a browser-based dictionary-attack tool, and the attack was launched against a built-in, 100-word wordlist.

The cracker worked through the wordlist in the browser, trying one candidate password after another, and eventually reported PASSWORD CRACKED SUCCESSFULLY along with the recovered password, password1. Opening PDF2 with that password confirmed it was correct and revealed the second captured flag, nw{networkwalks_persistence_jtr_270521}, which brought this exercise to a successful close.

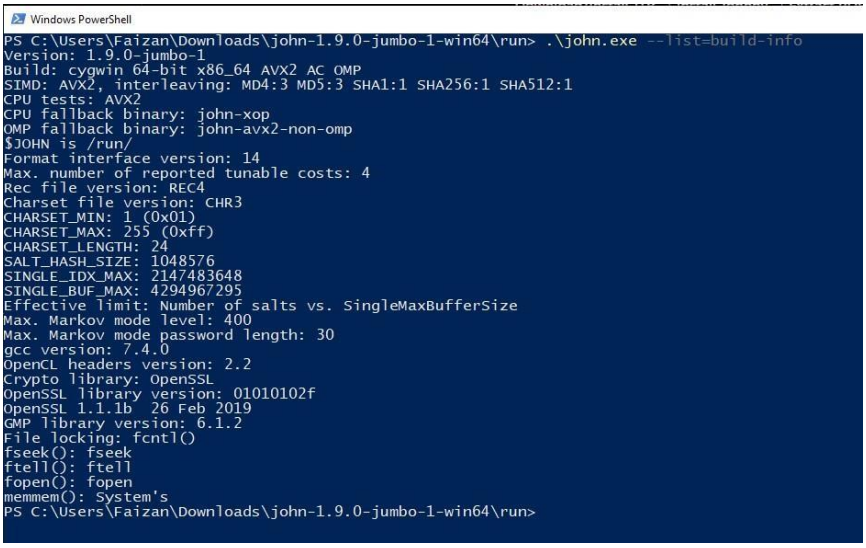
5. Overall Results

Both practical modules assigned for Week 3 were completed successfully. W3-PM1 was completed using John the Ripper together with its Johnny GUI, resulting in the successful recovery and verification of the PDF1 password. W3-PM2 was completed using the Networkwalks Hash Calculator and Password Cracker, resulting in the successful recovery of the PDF2 password, capture of its flag, and verification that the file opened correctly with the recovered credential.

6. Evidences Collected

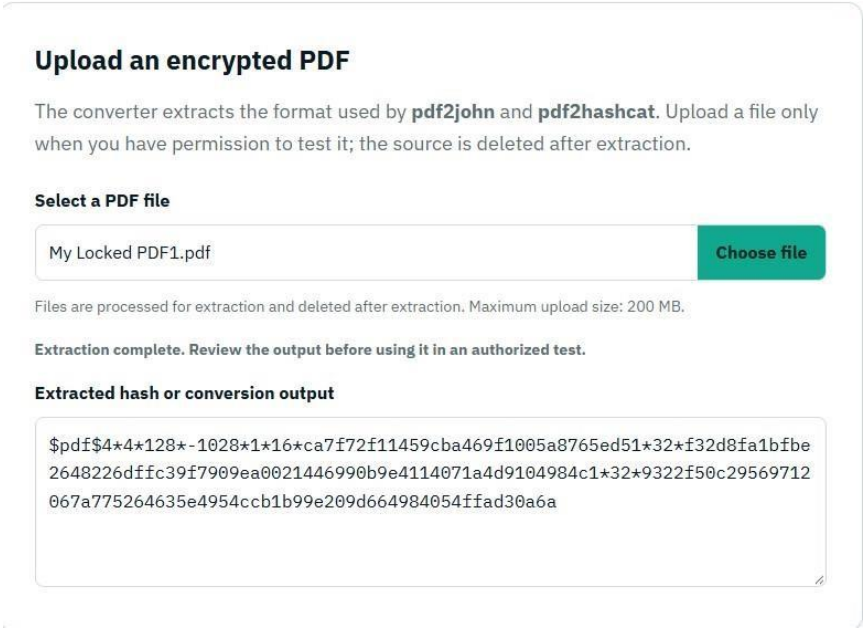
The screenshots below support the activities described in Section 4. Each one is followed by a short description of what it shows.

Screenshot 1: W3-PM1 — JTR build and version information



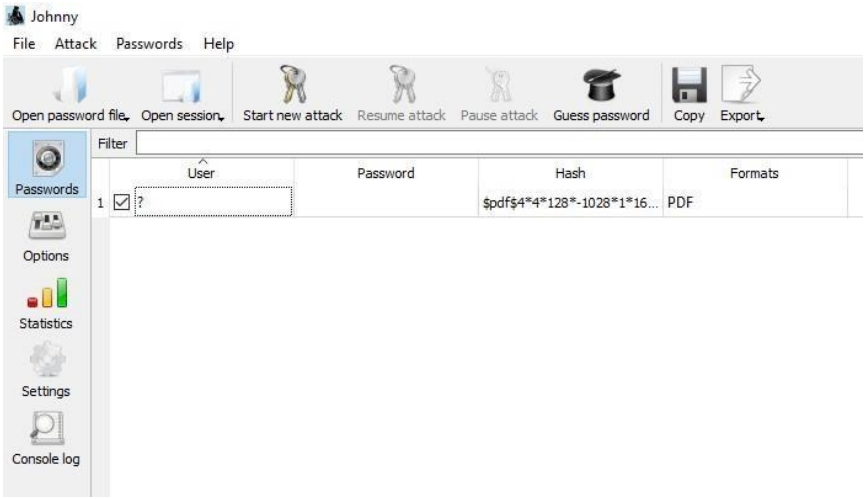
Description: *This screenshot captures the output of john.exe --list=build-info in PowerShell. It confirms the installed John the Ripper build, its supported hash formats and cryptographic libraries, which verifies that the tool was correctly installed before any cracking attempt.*

Screenshot 2: W3-PM1 — Hash extracted from PDF1



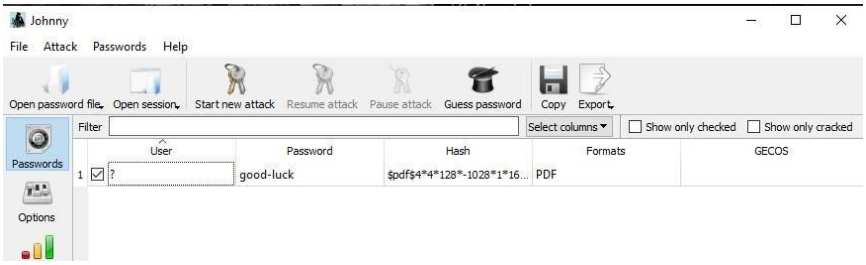
Description: *The Networkwalks Hash Calculator was used to upload My Locked PDF1.pdf and extract a crackable hash in pdf2john / hashcat format. This hash is the input that John the Ripper needs in order to attempt password recovery.*

Screenshot 3: W3-PM1 — PDF1 hash loaded into Johnny



Description: *The extracted pdf hash has been imported into the Johnny GUI, where it appears as a single PDF-format entry with no password yet recovered. This is the state of the attack just before it was started.*

Screenshot 4: W3-PM1 — Successful PDF1 password recovery



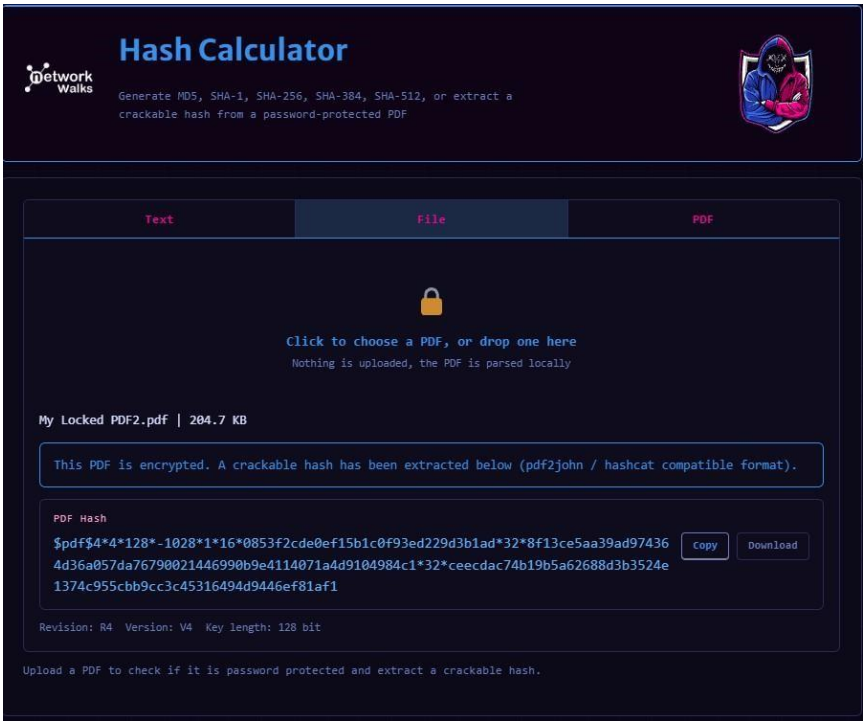
Description: *After running the attack, Johnny displays the recovered password good-luck next to the matching hash entry. This confirms that the dictionary attack against PDF1 succeeded.*

Screenshot 5: W3-PM1 — PDF1 successfully verified



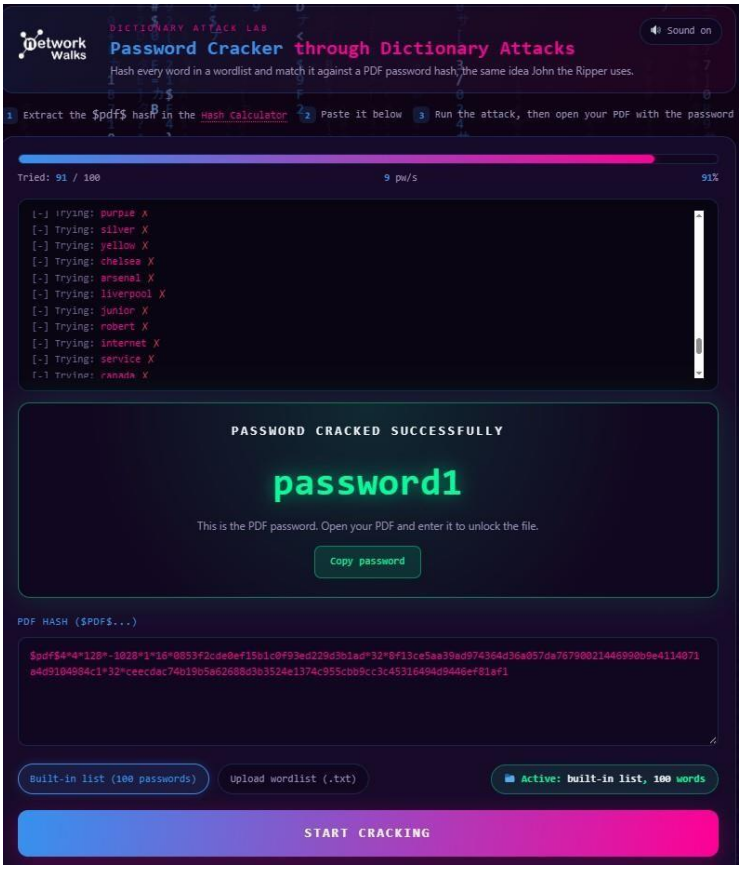
Description: *This is the Networkwalks confirmation screen shown after the recovered password was used to open PDF1. It displays the first captured flag, nw{cybersecurity_flag_captured_2608}, as proof that the file was successfully unlocked.*

Screenshot 6: W3-PM2 — Hash extracted from PDF2



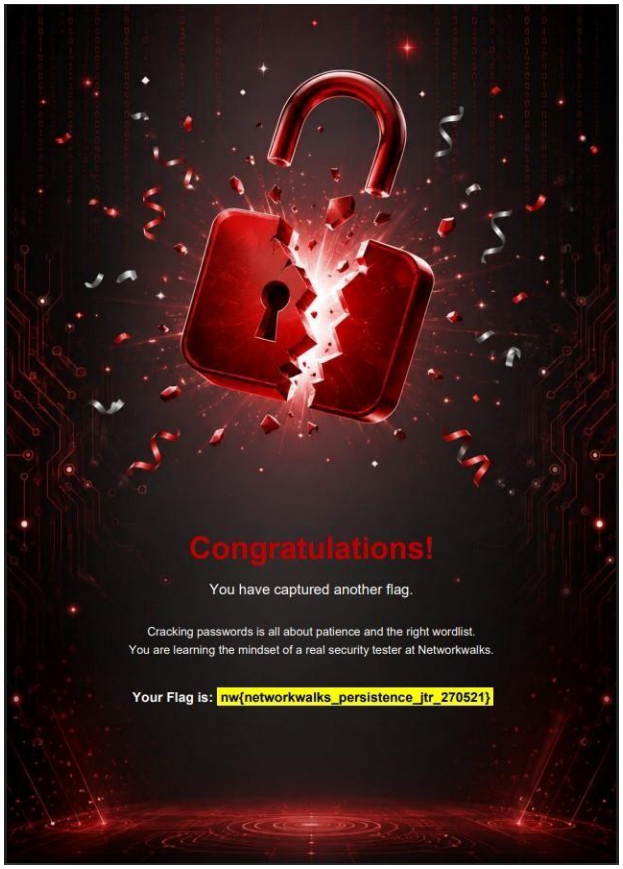
Description: *The Networkwalks Hash Calculator was used again, this time on My Locked PDF2.pdf, to produce a second crackable hash. The tool also reports the PDF's revision, version and key length, giving extra context on the encryption used.*

Screenshot 7: W3-PM2 — PDF2 password cracked via dictionary attack



Description: *The Networkwalks Password Cracker ran a dictionary attack straight in the browser, trying each word from a 100-word built-in wordlist against the PDF2 hash. The tool reports success and reveals the recovered password, password1.*

Screenshot 8: W3-PM2 — PDF2 successfully verified



Description: *This is the final confirmation screen shown once PDF2 was opened with the recovered password. It displays the second captured flag, nw{networkwalks_persistence_jtr_270521}, confirming the exercise was completed.*

7. Conclusion

The Week 3 project gave me hands-on experience with the full lifecycle of PDF password recovery: extracting a crackable hash from a protected file, running a dictionary attack against that hash with two different tools, and then verifying the recovered password by actually opening the file. Working through the exercise with both John the Ripper and the Networkwalks web tools let me compare a traditional command-line workflow against a more modern browser-based one, while reaching the same end goal in both cases.

Beyond the technical steps, the exercise reinforced the importance of clear evidence documentation — capturing each stage of the process in a screenshot so that the recovered password and the captured flag can both be traced back to the exact command or action that produced them. All of this work was completed strictly within the authorised educational scope defined for the lab.

8. Ethical Use Statement

The techniques documented in this report — hash extraction, dictionary attacks and password-protected PDF recovery — are intended strictly for authorised security testing, education and research. Running

these same techniques against a file, system or account without the explicit permission of its owner is unauthorised, and depending on the jurisdiction, may be illegal. Readers of this report are expected to apply these skills only within a scope they have been given clear permission to test.